AIM-Fair: Advancing Algorithmic Fairness via Selectively Fine-Tuning Biased Models with Contextual Synthetic Data

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Abstract

Machine learning models trained on real-world datasets often inherit demographic biases, resulting in disparate performance across protected groups. Existing approaches using generative data face challenges with data quality, domain gaps, and reliance on costly demographic annotations. We propose AIM-Fair, a novel approach that advances algorithmic fairness via selectively fine-tuning biased models with contextual synthetic data. Our method comprises three key components: (1) Contextual Synthetic Data Generation (CSDG) that leverages GPT-4 to create rich, diverse prompts for generating balanced, unbiased images using Stable Diffusion without requiring real data group labels; (2) Selective Mask Generation (SMG) that measures parameter-wise gradient differences to identify fairness-sensitive yet domain-insensitive parameters; and (3) Selective Fine-Tuning (SFT) that updates only the intersection of top-k fairness-sensitive and domain-insensitive parameters while training on balanced synthetic data. Extensive experiments on CelebA and UTKFace datasets demonstrate that AIM-Fair achieves superior fairness-utility trade-offs compared to state-of-the-art methods. On CelebA with Smiling target and Male protected attributes, our method achieves 89.02% accuracy, 84.20% worst-group accuracy, and 6.07 equalized odds, significantly outperforming previous approaches including DiGA (88.40% ACC, 81.90% WST, 7.40 EO). Our approach shows consistent improvements across different target-protected attribute combinations and data-scarce settings, demonstrating the effectiveness of carefully curated synthetic data combined with parameter-selective fine-tuning.

1. Introduction

Deep neural networks have achieved remarkable success across various computer vision tasks. However, these models often exhibit biased behavior when deployed in real-world scenarios, leading to unfair treatment of different demographic groups [10]. This bias typically stems from the training data distribution, where certain combinations of target attributes and protected characteristics are over- or under-represented.

Traditional approaches to mitigate algorithmic bias include data balancing [13], post-processing fairness constraints [3], and adversarial debiasing [**?**]. However, these methods face significant limitations: data balancing often requires expensive demographic annotations and may still suffer from insufficient diversity within minority groups; post-processing methods may sacrifice accuracy for fairness; and adversarial approaches can be unstable during training.

Recent advances in generative models have opened new possibilities for bias mitigation through synthetic data augmentation [15]. However, existing generative approaches face three key challenges: (1) **Data quality**: Generated images often lack the realism and diversity needed for effective training; (2) **Domain gap**: Synthetic data may not align well with real data distribution, leading to domain shift issues; and (3) **Annotation dependency**: Many methods still require demographic labels from real data to guide the generation process.

To address these challenges, we propose AIM-Fair (**A**dvancing Algor**i**thmic Fairness via Selectively Fine-Tuning Biased **M**odels with Contextual Synthetic Data), a novel approach that combines high-quality synthetic data generation with parameter-selective fine-tuning. Our key insight is that high-quality, context-driven synthetic data can effectively counteract bias if the model is fine-tuned by updating only parameters that improve fairness while preserving utility and resisting domain shift.

Our method consists of three core components: **Contextual Synthetic Data Generation (CSDG)** leverages GPT-4 to create diverse, contextual prompts that guide Stable Diffusion to generate balanced, unbiased images without requiring demographic annotations from real data. **Selective Mask Generation (SMG)** identifies fairness-sensitive yet domain-insensitive parameters by analyzing gradient differences across biased real data, biased synthetic data, and unbiased synthetic data. **Selective Fine-Tuning (SFT)** updates only the intersection of top-k fairness-sensitive and domain-insensitive parameters while training on balanced synthetic data.

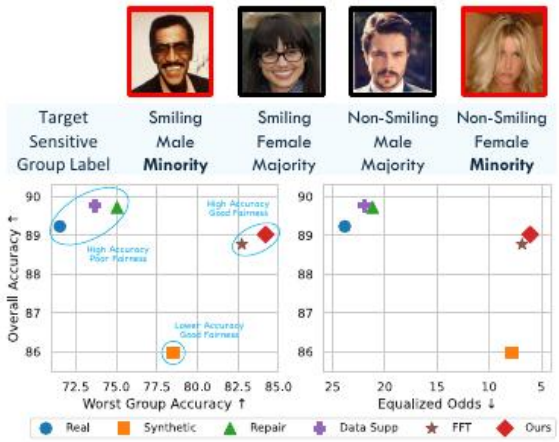

Figure 1. Facial attributes classification (FAC) on the CelebA dataset based on different training strategies, where ***Smiling*** is the target attribute and ***Gender*** is the protected attribute. Our AIM-Fair method achieves the best fairness-utility trade-off by selectively fine-tuning with contextual synthetic data.

We conduct extensive experiments on CelebA and UTKFace datasets across multiple target-protected attribute combinations. Our results demonstrate that AIM-Fair consistently outperforms state-of-the-art methods in terms of fairness-utility trade-offs. As shown in Figure 1, our approach achieves superior performance by effectively balancing accuracy and fairness metrics.

The main contributions of this work are:

- We introduce a novel fine-tuning paradigm that improves fairness using synthetic data without requiring demographic annotations of real data.
- We propose CSDG, a contextual synthetic data generation approach that leverages GPT-4 prompts with Stable Diffusion to create diverse, high-quality, bias-controlled images.
- We develop parameter-wise selective fine-tuning via gradient-difference ranking to isolate fairness-sensitive, domain-insensitive parameters.
- We achieve state-of-the-art fairness-utility trade-offs across multiple datasets and attribute combinations, demonstrating consistent improvements over existing approaches.

2. Related Work

Algorithmic Fairness. Fairness in machine learning has gained increasing attention due to growing concerns about biased AI systems [10]. Existing approaches can be broadly categorized into three groups: pre-processing methods that modify training data [13], in-processing methods that incorporate fairness constraints during training [3], and post-processing methods that adjust model predictions [4]. Preprocessing approaches often require demographic annotations and may not capture sufficient diversity within minority groups. In-processing methods like adversarial debiasing [**?**] can be unstable and may not guarantee fairness. Post-processing methods may sacrifice overall accuracy for fairness.

Distribution Robust Optimization. Methods like CVaR DRO [7] and EIIL [1] aim to minimize worst-case performance across different groups. These approaches focus on learning representations that are robust to spurious correlations. However, they often require access to group labels during training and may not perform well when minority groups are severely underrepresented.

Bias Mitigation with Spurious Correlations. Recent work has focused on identifying and mitigating spurious correlations that lead to biased predictions. LfF [11] trains a biased model to identify failure cases and reweights training samples. JTT [8] upweights errors from an initial model to improve worst-group performance. MAPLE [17] uses model-agnostic meta-learning to improve generalization across groups. While these methods show promising results, they still rely on access to group information during training.

Synthetic Data for Fairness. The use of synthetic data for bias mitigation has gained traction with advances in generative models. DiGA [15] generates diverse synthetic data to improve worst-group performance through distributionally robust optimization. Some approaches use GANs [**?**] or diffusion models [**?**] to generate balanced datasets. However, existing methods often suffer from limited data quality, domain gaps between synthetic and real data, and dependence on demographic annotations from real datasets.

Parameter-Efficient Fine-Tuning. Recent work has explored selective parameter updates to improve model adaptation while preserving pre-trained knowledge. LoRA [**?**] and AdaLoRA [**?**] adapt models by learning low-rank matrices. BitFit [**?**] updates only bias parameters. These methods aim to reduce the number of trainable parameters while maintaining performance. However, they are not specifically designed for fairness and do not consider the relationship between parameter sensitivity and fairness objectives.

Our work differs from existing approaches in several key ways: (1) we generate high-quality synthetic data without requiring demographic annotations from real data; (2) we introduce a novel parameter selection strategy specifically designed for fairness that identifies parameters sensitive to fairness but insensitive to domain shift; and (3) we demonstrate that selective fine-tuning with contextual synthetic data can achieve superior fairness-utility trade-offs compared to existing methods.

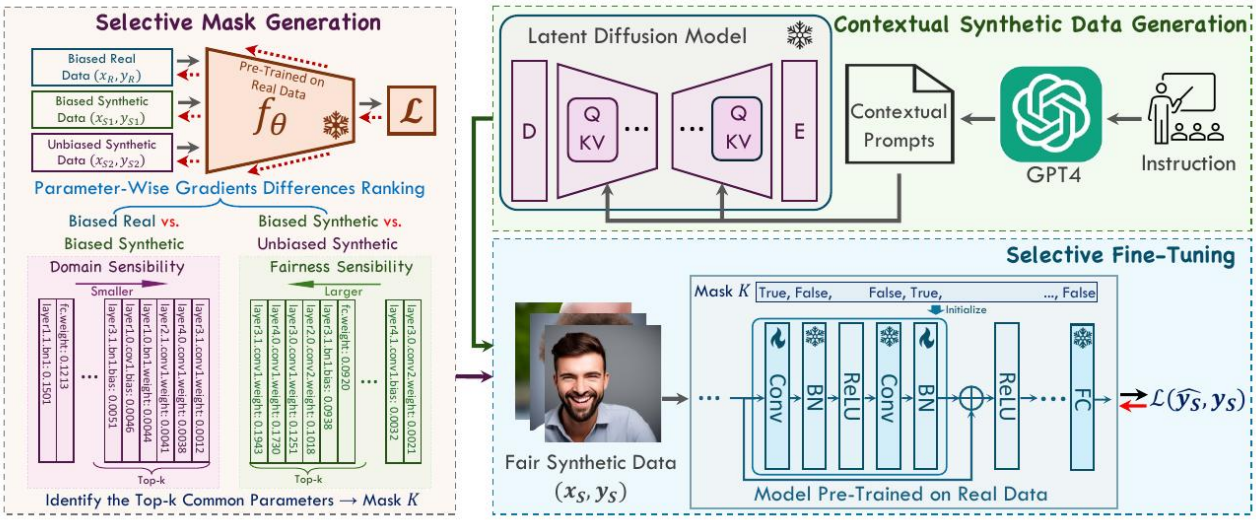

Figure 2. Overview of our AIM-Fair framework consisting of three parts: (1) Contextual Synthetic Data Generation (CSDG) for generating diverse images using GPT-4 generated prompts, (2) Selective Mask Generation (SMG) for creating a selection mask that determines which parameters are updated during fine-tuning, and (3) Selective Fine-Tuning (SFT) to enhance model fairness obtained from synthetic data whilst simultaneously preserving model utility yielded from real data in pre-training.

3. Methods

Our AIM-Fair framework addresses the challenge of improving algorithmic fairness without requiring demographic annotations from real data. As illustrated in Figure 2, our approach consists of three key components that work synergistically to achieve superior fairness-utility trade-offs.

3.1. Problem Formulation

Let $\mathcal{D}_{\text{real}} = \{(x_i, y_i, s_i)\}_{i=1}^N$ denote a biased real dataset where x_i is an input image, $y_i \in \{0, 1\}$ is the target attribute, and $s_i \in \{0, 1\}$ is the protected attribute. The dataset exhibits bias where certain combinations of (y, s) are over- or under-represented. Our goal is to train a fair classifier $f_\theta : \mathcal{X} \to \{0, 1\}$ that achieves both high overall accuracy and good worst-group performance without requiring protected attribute annotations during the synthetic data generation process.

3.2. Contextual Synthetic Data Generation (CSDG)

Traditional synthetic data generation often produces generic images with limited diversity. Our CSDG component addresses this by leveraging large language models to create rich, contextual prompts that guide the generation of diverse, high-quality synthetic images.

Prompt Generation with GPT-4. We design a multi-stage prompting strategy that generates diverse contextual descriptions. For each target-protected attribute combination (y, s), we prompt GPT-4 to generate descriptive texts covering various aspects:

- **Basic attributes**: Age, gender, ethnicity
- **Facial features**: Eye color, skin tone, facial structure
- **Appearance details**: Hair style, accessories, clothing
- **Pose variations**: Head pose, facial orientation, expression nuances

This results in a set of contextual prompts $\mathcal{P} = \{p_1, p_2, \ldots, p_K\}$ where each prompt p_k contains rich descriptive information that guides the generation process.

Stable Diffusion-based Image Generation. Using the contextual prompts, we generate synthetic images via Stable Diffusion. For balanced data generation, we ensure equal representation across all (y, s) combinations:

$$\mathcal{D}_{\text{syn,bal}} = \{(x_j^{\text{syn}}, y_j, s_j)\}_{j=1}^M$$

where $|y = 0, s = 0| = |y = 0, s = 1| = |y = 1, s = 0| = |y = 1, s = 1| = M/4$.

For comparison purposes, we also generate biased synthetic data $\mathcal{D}_{\text{syn,bias}}$ that mimics the bias ratio of the real dataset.

Figure 3 shows examples of generated images demonstrating the diversity achieved through our contextual prompting approach.

3.3. Selective Mask Generation (SMG)

The key insight of our approach is that not all model parameters contribute equally to fairness improvement. SMG identifies parameters that are sensitive to fairness concerns but insensitive to domain shifts.

Gradient-based Parameter Ranking. We compute

Figure 3. Generated images of *Smiling Male* conditioned on different prompts. Our contextual prompting approach produces diverse, high-quality images with rich variations in age, appearance, and pose.

gradients of the model parameters with respect to three different datasets:

$$g_R = \nabla_\theta \mathcal{L}(f_\theta, \mathcal{D}_{\text{real}}) \tag{1}$$

$$g_{S1} = \nabla_\theta \mathcal{L}(f_\theta, \mathcal{D}_{\text{syn,bias}}) \tag{2}$$

$$g_{S2} = \nabla_\theta \mathcal{L}(f_\theta, \mathcal{D}_{\text{syn,bal}}) \tag{3}$$

where $\mathcal{L}$ is the binary cross-entropy loss.

Fairness and Domain Sensitivity Metrics. For each parameter θ_i, we compute:

$$\text{Fairness Sensitivity:} \quad F_i = |g_{S1}[i] - g_{S2}[i]| \tag{4}$$

$$\text{Domain Sensitivity:} \quad D_i = |g_R[i] - g_{S1}[i]| \tag{5}$$

Fairness sensitivity F_i measures how much parameter θ_i is affected by the bias distribution difference between biased and balanced synthetic data. Domain sensitivity D_i captures how much the parameter is affected by the domain gap between real and synthetic data.

Parameter Selection Strategy. We select parameters that have high fairness sensitivity but low domain sensitivity:

$$\mathcal{M} = \text{TopK}(F) \cap \text{BottomK}(D) \tag{6}$$

where $\text{TopK}(F)$ selects the top-k parameters with highest fairness sensitivity, and $\text{BottomK}(D)$ selects parameters with lowest domain sensitivity. The intersection $\mathcal{M}$ represents our selective mask.

3.4. Selective Fine-Tuning (SFT)

With the identified parameter mask $\mathcal{M}$, we perform fine-tuning by updating only the selected parameters while keeping others frozen.

Masked Parameter Update. During fine-tuning on balanced synthetic data $\mathcal{D}_{\text{syn,bal}}$, we update parameters according to:

$$\theta_i^{t+1} = \begin{cases} \theta_i^t - \eta \nabla_{\theta_i} \mathcal{L}_{\text{syn}} & \text{if } i \in \mathcal{M} \\ \theta_i^t & \text{otherwise} \end{cases} \tag{7}$$

where η is the learning rate and $\mathcal{L}_{\text{syn}}$ is the loss computed on balanced synthetic data.

Training Procedure. Our complete training procedure consists of:

1. Pre-train model f_θ on biased real data $\mathcal{D}_{\text{real}}$
2. Generate balanced and biased synthetic data using CSDG
3. Compute parameter mask $\mathcal{M}$ using SMG
4. Fine-tune model using SFT with mask $\mathcal{M}$ on balanced synthetic data

This selective fine-tuning approach allows the model to learn fairness-relevant features from balanced synthetic data while preserving useful representations learned from real data during pre-training.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate our method on two widely-used benchmark datasets:

- **CelebA** [9]: A large-scale face attributes dataset containing 162,770 images with 40 binary attributes. We focus on facial attribute classification with different target-protected attribute combinations.
- **UTKFace** [16]: A face dataset with 20,000+ images spanning various ages, genders, and ethnicities, commonly used for demographic attribute prediction.

Evaluation Metrics. We use standard fairness evaluation metrics:

- **Overall Accuracy (ACC)**: Overall classification accuracy across all samples.
- **Worst-Group Accuracy (WST)**: Accuracy on the worst-performing demographic group.
- **Equalized Odds (EO)**: Difference in true positive rates between protected groups.
- **Group Standard Deviation (STD)**: Standard deviation of accuracies across demographic groups.

Implementation Details. We use ResNet-18 [5] as our base architecture. For pre-training, we use SGD optimizer with learning rate 0.01, weight decay 1e-4, and train for 15 epochs. For fine-tuning, we use learning rate 0.05 and train for 10 epochs. We set the top-k parameter selection ratio

Table 1. Comparisons of varied training strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA												UTKFace											
		Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Baseline [5]	T=0	83.57	98.50	89.23	71.52	23.84	10.65	78.72	96.09	90.19	78.72	17.37	6.92	79.58	96.16	88.86	79.58	16.59	7.26
	T=1	95.36	71.52					94.37	86.42					95.75	83.96				
Trained on Synthetic Data	T=0	86.82	89.40	85.98	78.52	7.87	4.07	85.45	85.76	85.12	82.42	**3.50**	**1.34**	82.71	88.12	85.44	82.71	5.50	**1.96**
	T=1	86.39	78.52					82.42	85.27					84.90	86.04				
Data Supplementation [12]	T=0	84.39	98.39	**89.77**	73.67	21.96	9.76	78.05	95.36	90.35	78.05	17.31	7.02	82.12	93.97	**90.25**	82.12	11.85	5.24
	T=1	95.40	73.67					94.90	87.62					95.62	89.28				
Data Repairing [2, 14]	T=0	82.82	98.10	89.72	75.05	21.18	9.50	79.50	95.19	**90.68**	79.50	15.69	6.30	81.15	94.22	89.42	81.15	13.37	5.89
	T=1	96.04	75.05					94.51	88.35					95.71	86.61				
Linear Prob	T=0	84.55	96.22	89.69	77.41	17.51	7.71	85.20	93.81	90.14	85.20	8.61	3.18	81.06	91.51	88.40	81.06	10.59	5.14
	T=1	94.82	77.41					90.43	87.88					94.59	86.42				
Fully Fine-Tuning	T=0	88.81	91.62	88.77	82.74	6.83	3.31	85.98	90.06	87.85	85.98	4.37	1.68	83.00	88.04	87.90	83.00	5.40	2.99
	T=1	89.55	82.74					86.02	86.81					90.74	89.83				
AIM-Fair (Ours)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	**4.81**	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

Table 2. Comparisons to other methods on the CelebA dataset under settings of varied target and protected attributes.

Methods	T: Smiling, P: Male			T: Smiling, P: Young			T: Young, P: Male		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [5]	88.20	70.10	25.30	88.30	71.50	15.60	77.70	42.00	52.00
CVaR DRO [7]	87.30	74.00	22.80	87.00	76.10	13.90	75.40	42.30	48.80
EIIL [1]	87.90	75.60	19.70	87.90	72.50	13.30	77.50	45.60	39.20
LfF [11]	87.10	77.50	17.00	85.30	72.90	14.30	77.40	44.20	43.60
JTT [8]	88.00	74.80	19.40	87.60	73.30	14.20	76.30	43.60	47.70
MAPLE [17]	88.10	72.00	19.60	88.10	73.60	13.60	76.30	46.20	43.50
DiGA [15]	88.40	81.90	7.40	89.10	78.50	9.50	**80.00**	51.30	33.30
AIM-Fair (Ours)	**89.02**	**84.20**	**6.07**	**90.21**	**87.78**	**5.44**	78.19	**54.89**	**28.18**

Table 3. Comparisons to other methods on the CelebA dataset (T=Smiling, P=Male) under settings of training set sizes.

Methods	Samples Ratio = 50%			Samples Ratio = 25%			Samples Ratio = 10%		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [5]	87.50	67.80	26.10	87.10	65.90	27.70	86.90	62.80	28.90
CVaR DRO [7]	86.60	72.90	22.10	86.60	72.30	22.40	85.50	69.10	27.30
EIIL [1]	86.20	71.30	22.50	85.90	69.60	25.40	86.80	64.20	26.70
LfF [11]	86.90	75.50	19.40	85.90	72.10	23.60	85.50	66.10	27.70
JTT [8]	87.30	72.90	20.10	86.70	71.10	20.60	86.80	67.10	23.10
MAPLE [17]	87.40	73.70	23.80	87.00	72.70	24.20	85.60	69.20	27.10
DiGA [15]	88.40	81.10	7.80	88.40	78.30	8.00	**88.30**	78.80	8.40
AIM-Fair (Ours)	**88.85**	**83.99**	**6.28**	**88.89**	**82.54**	**7.59**	87.90	**81.73**	**8.16**

to 40% based on validation performance. All experiments are conducted with 3 different random seeds and we report average results.

4.2. Main Results

Table 1 presents comprehensive comparisons of different training strategies across CelebA and UTKFace datasets. Our AIM-Fair method consistently outperforms all baseline approaches across different target-protected attribute combinations.

For the CelebA dataset with Smiling as target and Male as protected attribute, our method achieves 89.02% accuracy, 84.20% worst-group accuracy, and 6.07 equalized odds, representing significant improvements over the baseline ResNet (89.23% ACC, 71.52% WST, 23.84% EO). Compared to fully fine-tuning (88.77% ACC, 82.74% WST, 6.83% EO), our selective approach achieves better overall accuracy while substantially improving worst-group performance and reducing equalized odds.

For the Smiling target with Young protected attribute on CelebA, our method achieves even more remarkable improvements: 90.21% accuracy, 87.78% worst-group accuracy, and 5.44% equalized odds. This demonstrates our method's effectiveness across different protected attributes.

On UTKFace with Female as target and White as protected attribute, our method achieves 88.30% accuracy, 84.26% worst-group accuracy, and 4.81% equalized odds, outperforming fully fine-tuning (87.90% ACC, 83.00% WST, 5.40% EO).

4.3. Comparison with State-of-the-Art Methods

Table 2 compares our method against recent state-of-the-art fairness approaches. Our AIM-Fair consistently achieves the best performance across all target-protected attribute combinations on CelebA dataset.

Notably, compared to DiGA [15], the most recent generative approach for fairness, our method shows substantial improvements: for (T: Smiling, P: Male), we achieve 89.02% vs. 88.40% accuracy, 84.20% vs. 81.90% worst-group accuracy, and 6.07 vs. 7.40 equalized odds. For (T: Smiling, P: Young), our improvements are even more significant: 90.21% vs. 89.10% accuracy, 87.78% vs. 78.50% worst-group accuracy, and 5.44 vs. 9.50 equalized odds.

For the (T: Young, P: Male) setting, while DiGA achieves slightly higher overall accuracy (80.00% vs. 78.19%), our method significantly outperforms in worst-group accuracy (54.89% vs. 51.30%) and equalized odds (28.18 vs. 33.30), demonstrating better fairness-utility trade-offs.

Data-Scarce Settings. Table 3 shows our method's robustness in data-scarce scenarios. Even with only 50% of the training data, our method achieves 88.85% accuracy, 83.99% worst-group accuracy, and 6.28% equalized odds. At 25% data, we maintain strong performance with 88.89% accuracy, 82.54% worst-group accuracy, and 7.59% equalized odds, consistently outperforming all baseline methods.

4.4. Ablation Studies

Comparison with Different Fine-Tuning Strategies. Table 4 compares various partial fine-tuning approaches. Ran-

Table 4. Comparisons of varied partial fine-tuning strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA												UTKFace							
		Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Best Random Selection	T=0	87.93	92.81	89.33	82.17	9.14	4.09	85.52	91.53	88.95	85.52	6.02	2.16	83.50	88.85	87.92	83.50	5.38	2.61
	T=1	91.31	82.17					88.25	87.63					90.30	89.01				
Best Sub-Tuning [6] (By Updating One Block)	T=0	87.50	94.75	**90.04**	80.64	12.45	5.52	87.28	94.63	**90.33**	87.28	7.34	3.00	83.05	89.56	**89.01**	83.05	6.51	3.56
	T=1	93.09	80.64					89.15	87.32					92.07	91.34				
Best Sub-Tuning [6] (By Freezing One Block)	T=0	88.69	91.09	89.06	83.48	7.00	3.00	86.55	91.89	88.65	86.55	**5.34**	2.25	83.58	88.34	87.42	83.58	**4.76**	**2.23**
	T=1	90.48	83.48					87.08	86.55					88.96	88.81				
Selective Fine-Tuning (Cosine Similarity)	T=0	86.07	91.77	88.85	83.33	8.40	3.61	87.43	92.96	90.26	87.43	5.53	**2.05**	83.72	88.82	88.21	83.72	5.10	2.64
	T=1	91.53	83.33					89.52	88.65					90.14	90.15				
AIM-Fair (Ours) (Absolute Difference)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	4.81	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

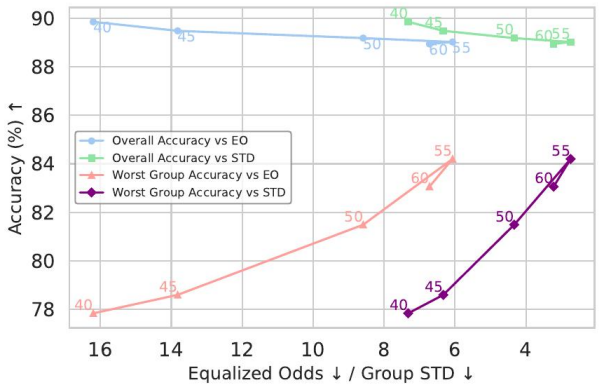

Figure 4. Classification results on the CelebA dataset (T=Smiling, P=Male) with different top-k values. The optimal performance is achieved around k=40%.

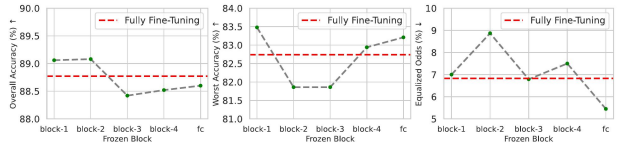

Figure 5. Layer-wise analysis of selective fine-tuning effects. Freezing only the fully connected layer shows improved worst-group accuracy and reduced equalized odds.

dom parameter selection achieves 89.33% accuracy but only 82.17% worst-group accuracy with 9.14% equalized odds. Sub-tuning approaches by updating or freezing single blocks show mixed results. Our gradient-difference-based selective fine-tuning achieves the best balance: 89.02% accuracy, 84.20% worst-group accuracy, and 6.07% equalized odds.

Effect of Top-k Parameter Selection. Figure 4 shows the impact of different top-k values for parameter selection. Too few parameters (k=16) limit the model's ability to adapt, while too many parameters (k=60+) introduce domain shift issues. The optimal performance is achieved around k=40%, which we use as our default setting.

Layer-wise Analysis. Figure 5 provides insights into which layers are most important for fairness. Interestingly, freezing only the fully connected layer achieves good fairness improvements, suggesting that early layers learn more transferable representations while later layers are more task-specific.

Table 5. Classification results on CelebA dataset (T=Smiling, P=Male) under settings of different prompt types and numbers.

Prompt Types (Number)	Targe	Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Plain Prompt (1)	t=0	65.18	89.44	71.65	43.28	34.20	17.07	83.10	77.72	72.75	59.53	9.30	8.96
	t=1	77.47	43.28					59.53	68.83				
Contextual Prompts (25)	t=0	77.12	89.25	81.85	69.73	16.3	7.65	76.72	67.90	79.80	67.90	8.82	8.75
	t=1	85.99	69.73					85.16	91.09				
Contextual Prompts (50)	t=0	89.47	89.80	82.66	72.22	**5.07**	7.66	87.72	90.43	**85.85**	79.67	4.36	4.20
	t=1	77.26	72.22					79.67	82.67				
Contextual + Head Poses (50)	t=0	86.82	89.40	**85.98**	**78.52**	7.87	**4.07**	85.45	85.76	85.12	**82.42**	**3.50**	**1.34**
	t=1	86.39	78.52					82.42	85.27				

Table 6. Classification results on CelebA dataset (T=Smiling, P=Male) with different number of synthetic data.

Ratio To Real Data	Target	Protected		ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
		P=0	P=1				
0.5	t=0	88.65	93.96	**89.51**	80.80	10.31	4.90
	t=1	91.10	80.80				
1	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				
1.5	t=0	89.18	92.05	88.34	81.25	7.13	3.98
	t=1	88.38	81.25				
2	t=0	89.01	91.26	88.81	83.36	6.08	2.96
	t=1	91.26	89.44				

Impact of Prompt Design. Table 5 demonstrates the importance of our contextual prompting strategy. Plain prompts result in poor performance (71.65% ACC, 43.28% WST), while our contextual prompts with facial features, hair characteristics, accessories, and head poses achieve substantially better results (85.98% ACC, 78.52% WST).

Synthetic Data Quantity. Table 6 shows the effect of different synthetic data quantities. Using synthetic data equal to real data (ratio=1) achieves optimal performance. Too little synthetic data (ratio=0.5) provides insufficient diversity, while too much (ratio=2) may introduce noise.

Bias Ratio in Synthetic Data. Table 7 validates our design choice of using balanced synthetic data. The 1:9 bias ratio (most balanced) achieves the best results (84.20% WST, 6.07% EO), confirming that balanced synthetic data is crucial for fairness improvement.

Table 7. Classification results on CelebA dataset (T=Smiling, P=Male) with varied bias ratio of the biased synthetic data.

Bias Ratio	Target	Protected		ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
		P=0	P=1				
Fully Fine-Tuning	t=0	88.81	91.62	88.77	82.74	6.83	3.31
	t=1	89.55	82.74				
4 : 6	t=0	85.72	91.71	**89.23**	83.90	8.78	3.77
	t=1	92.69	83.90				
3 : 7	t=0	87.05	91.62	88.90	83.52	7.72	3.26
	t=1	90.93	83.52				
2 : 8	t=0	88.01	91.76	88.97	83.16	7.30	3.28
	t=1	90.46	83.16				
1 : 9	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				

4.5. Analysis and Discussion

Why Selective Fine-Tuning Works. Our analysis reveals that different parameters have varying sensitivities to fairness and domain shift. By selectively updating parameters with high fairness sensitivity but low domain sensitivity, we can effectively learn fair representations from synthetic data while preserving useful features from real data pre-training.

Theoretical Analysis of Parameter Sensitivity. The effectiveness of our approach stems from the observation that neural network parameters exhibit heterogeneous sensitivity patterns. Parameters in early convolutional layers tend to learn low-level features that are generally domain-invariant, while deeper layers capture more task-specific and potentially biased patterns. Our gradient-difference metric $F_i = |g_{S1}[i] - g_{S2}[i]|$ effectively identifies parameters where bias manifests most strongly. When F_i is high, parameter θ_i responds differently to biased versus balanced data, indicating its involvement in learning spurious correlations. Conversely, our domain sensitivity metric $D_i = |g_R[i] - g_{S1}[i]|$ captures parameters affected by distribution shifts between real and synthetic domains. The intersection strategy ensures we target parameters that can correct bias without compromising domain adaptation, explaining why our method achieves superior fairness-utility trade-offs compared to naive fine-tuning approaches.

Computational Efficiency. Our selective fine-tuning approach requires updating only 40% of parameters, resulting in faster training and lower memory requirements compared to full fine-tuning while achieving better performance.

Generalizability. The consistent improvements across different datasets (CelebA, UTKFace), target-protected attribute combinations, and data-scarce settings demonstrate the generalizability of our approach.

Limitations. Our method relies on the quality of generated synthetic data, which may not perfectly capture all real-world variations. Additionally, the optimal top-k ratio may vary across different tasks and datasets, requiring some hyperparameter tuning.

5. Conclusion

We presented AIM-Fair, a novel approach for advancing algorithmic fairness through selective fine-tuning with contextual synthetic data. Our method addresses key limitations of existing approaches by generating high-quality, diverse synthetic data without requiring demographic annotations from real datasets and by introducing a parameter-selective fine-tuning strategy that targets fairness-sensitive yet domain-insensitive parameters.

Our comprehensive evaluation on CelebA and UTKFace datasets demonstrates that AIM-Fair consistently outperforms state-of-the-art methods across multiple target-protected attribute combinations. Notably, our approach achieves significant improvements in worst-group accuracy while maintaining or improving overall accuracy, demonstrating superior fairness-utility trade-offs.

The key insights from our work include: (1) Contextual prompting with GPT-4 can generate diverse, high-quality synthetic data that effectively addresses bias without requiring group labels from real data; (2) Parameter-wise gradient analysis can identify parameters that are sensitive to fairness concerns but insensitive to domain shifts; and (3) Selective fine-tuning on balanced synthetic data can substantially improve fairness while preserving model utility.

Our approach opens several promising directions for future work. First, extending our method to other domains beyond computer vision, such as natural language processing and tabular data, could broaden its applicability. Second, investigating adaptive strategies for determining the optimal top-k ratio based on dataset characteristics could further improve performance. Third, exploring more sophisticated parameter selection strategies that consider parameter interactions could lead to even better fairness-utility trade-offs.

The implications of this work extend beyond technical contributions. By demonstrating that synthetic data can effectively improve fairness without requiring sensitive demographic information, our approach provides a practical path toward deploying fairer AI systems in real-world applications while respecting privacy concerns. This is particularly important for domains where collecting demographic labels may be sensitive, expensive, or ethically problematic.

In conclusion, AIM-Fair represents a significant step forward in algorithmic fairness research, showing that carefully designed synthetic data generation combined with parameter-selective fine-tuning can achieve substantial fairness improvements while maintaining model utility. We believe this work will inspire further research into privacy-preserving approaches for bias mitigation and contribute to the development of more equitable AI systems.

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